```
========================================
     SMART TRAFFIC MANAGEMENT SYSTEM
========================================
1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 1

Enter vehicle number: AP12TS1212
Enter vehicle type (car/bike/bus/ambulance/fireTruck): car
Enter road number (1-4): 1
Vehicle added successfully!
```

```
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9. Shortest Path
10. Exit

Enter your choice: 1

Enter vehicle number: AP11TS1111
Enter vehicle type (car/bike/bus/ambulance/fireTruck): ambulance
Enter road number (1-4): 1
Vehicle added successfully!
Emergency vehicle detected. Priority given.

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     SMART TRAFFIC MANAGEMENT SYSTEM
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4. Dequeue Vehicle
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7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 1
```

```
Enter vehicle number: TS12AM1212
Enter vehicle type (car/bike/bus/ambulance/fireTruck): bike
Enter road number (1-4): 2
Vehicle added successfully!

========================================
     SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
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4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 1

Enter vehicle number: TS12TS2341
Enter vehicle type (car/bike/bus/ambulance/fireTruck): bux
Enter road number (1-4): 2
Vehicle added successfully!
```

```
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1. Vehicle Entry
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7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 2

========== VEHICLE DETAILS ==========

Vehicle 1
Number : AP12TS1212
Type   : car
Road   : 1
Priority: NORMAL

Vehicle 2
Number : AP11TS1111
Type   : ambulance
Road   : 1
Priority: EMERGENCY

Vehicle 3
Number : TS12AM1212
Type   : bike
Road   : 2
Priority: NORMAL

Vehicle 4
Number : TS12TS2341
Type   : bux
Road   : 2
Priority: NORMAL
```

```
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9. Shortest Path
10. Exit

Enter your choice: 3

========== TRAFFIC QUEUES ==========

Road 1:
AP12TS1212 - car
AP11TS1111 - ambulance [EMERGENCY]

Road 2:
TS12AM1212 - bike
TS12TS2341 - bux

Road 3:
No vehicles

Road 4:
No vehicles
```

```
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     SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
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8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 4

Enter road number: 2
Vehicle TS12AM1212 has passed from Road 2.

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1. Vehicle Entry
2. Display Vehicles
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4. Dequeue Vehicle
5. Traffic Density Analysis
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7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 5

========== TRAFFIC DENSITY ==========
Road 1: 2 vehicles - LOW
Road 2: 1 vehicles - LOW
Road 3: 0 vehicles - LOW
Road 4: 0 vehicles - LOW
```

```
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     SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 6

========== SMART SIGNAL ==========

1. Road 1 -> EMERGENCY PRIORITY
   Vehicles: 2
   Green Light: 20 seconds

2. Road 2
   Vehicles: 1
   Green Light: 20 seconds

3. Road 3
   Vehicles: 0
   Green Light: 20 seconds

4. Road 4
   Vehicles: 0
   Green Light: 20 seconds
```

```
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     SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
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7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 7

========== EMERGENCY PRIORITY ==========
Emergency vehicle AP11TS1111 detected on Road 1
Road 1 gets PRIORITY.

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     SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 8
```

```
========== TRAFFIC REPORT ==========
Total vehicles: 3
Vehicles on Road 1: 2
Vehicles on Road 2: 1
Vehicles on Road 3: 0
Vehicles on Road 4: 0
Highest-density road: Road 1
Emergency vehicles: 1

Signal Timings:
Road 1: 20 seconds
Road 2: 20 seconds
Road 3: 20 seconds
Road 4: 20 seconds

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      SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 9

Enter source junction (1-4): 1
Enter destination junction (1-4): 4

Shortest distance: 6 units
Shortest Path: 1 -> 3 -> 4
```

```
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      SMART TRAFFIC MANAGEMENT SYSTEM
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1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 10
```